

1

Raw IMS vibration data

data/raw/1st_test, 2nd_test, and 3rd_test



2

Data validation

Snapshot counts, timestamp order, and channel-to-bearing mapping



3

Feature extraction

RMS, kurtosis, crest factor, and envelope fault-frequency energy



4

RUL labeling and preprocessing

Healthy-baseline normalization and normalized RUL target



5

PCA-HI analysis

Degradation consistency check from engineered vibration features



6

Model comparison

Data-only FNN, DeepXDE PINN, LSTM, and 1D CNN



7

Evaluation and reporting

MAE, RMSE, R-squared, rankings, and prediction curves